

GPLS

Gold Price Lookup System

Software Requirements Specification — Complete Edition

Document Version	SRS v2.0 — Full Edition
Date	May 2026
Status	Draft — Pending Review
Based on SRS	v1.0 — Nhóm 14 (March 2026)
Core Modules	8 modules (M01–M08)
New Features	11 features (F01–F11)
Total Releases	R1 (Q2/2026) · R2 (Q3/2026) · R3 (Q4/2026)

PART 1 — CORE FEATURES (SRS v1.0)

These modules are defined in the original SRS v1.0 (Nhóm 14, March 2026) and are required for Release 1 unless marked otherwise. All FR and NFR from the original document are preserved here in full.

M01 — Domestic Gold Price Display — Hiển thị giá vàng trong nước

The system crawls and displays domestic gold prices from SJC and DOJI (R1), PNJ and Bảo Tín Minh Châu (R2), with automatic anomaly detection and status labelling.

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-01.1	System must fetch and display buy/sell prices for SJC and DOJI	Must	Price table shows correct brand, gold type, and both buy/sell prices
FR-01.2	System must auto-refresh price data every 5 minutes during trading hours (07:00–17:00)	Must	Crawl log records correct cycle; no delay beyond 5 minutes
FR-01.3	Outside trading hours, system must display last known price with a clear timestamp	Must	Timestamp label "Updated at HH:mm DD/MM/YYYY" is always visible
FR-01.4	System must show status badge: Live (≤5 min, green) / Recent (5–30 min, yellow) / Outdated (>30 min, red)	Must	Badge colour matches rule; auto-updates as time passes without page reload
FR-01.5	System must show % price change vs previous update with ▲ / ▼ / – indicator	Must	Formula: $(\text{currentPrice} - \text{prevPrice}) / \text{prevPrice} \times 100$; green/red/grey colouring
FR-01.6	System must detect anomalous data (deviation > 15%) and suppress it from display	Must	Flagged records hidden from price table; Admin receives notification
FR-01.7	System must display all supported gold types: MIEN_SJC, NHAN_9999, VANG_24K, VANG_18K	Should	All active gold types of each brand appear in the table
FR-01.8	System must add PNJ and Bảo Tín Minh Châu to the price table (R2)	Could	R2: PNJ crawler operational; data appears in table

M02 — International Gold Price Display — Hiển thị giá vàng thế giới

Fetches XAU/USD from international APIs (Kitco, Gold API) and converts to VND/tael.

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-02.1	System must fetch XAU/USD from international API and display in real time	Must	<i>Data refreshes every 5 minutes; displayed in USD/troy ounce</i>
FR-02.2	System must convert international gold price to VND/tael	Must	<i>Formula: $\text{spotPriceUSD} \times \text{exchangeRateUSD_VND} \times (1/26.167 \times 1,000,000)$</i>
FR-02.3	System must support XAU/EUR display (R2)	Could	<i>R2: XAU/EUR currency pair added to international price table</i>

M03 — Price History Charts — Biểu đồ lịch sử giá

Displays line charts of gold price movements across selectable time ranges to support trend analysis.

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-03.1	System must display a line chart of historical gold prices	Must	Chart renders correct X-axis (time) and Y-axis (VND price)
FR-03.2	System must support time ranges: 1D, 1W, 1M (R1)	Must	Tab switch loads correct data from DB; load time ≤ 2 seconds
FR-03.3	System must support additional ranges: 3M, 1Y (R2)	Could	R2: two new selector options; full data available
FR-03.4	Users must be able to hover on chart points to see exact price and timestamp	Must	Tooltip shows correct price and timestamp at hovered point
FR-03.5	Users must be able to filter chart by brand and gold type	Should	Brand dropdown triggers chart data reload for selected combination
FR-03.6	When data is insufficient for selected range, system shows available data with a note	Must	Note "Data from [start date]" appears below chart
FR-03.7	Registered Users may export historical data to CSV (R3)	Could	R3: Export CSV button downloads file with timestamp, buy, sell columns

M04 — Brand Comparison Table — Bảng so sánh thương hiệu

Displays and compares gold prices across all active brands at the same point in time.

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-04.1	System must display a side-by-side buy/sell comparison table for all active brands	Must	All active brands listed; data from the same crawl session
FR-04.2	System must automatically highlight the brand with the highest buy price	Must	Highest buy-price cell has colour highlight; tooltip explains meaning
FR-04.3	System must automatically highlight the brand with the lowest sell price	Must	Lowest sell-price cell has colour highlight; tooltip explains meaning
FR-04.4	Users must be able to filter the comparison table by gold type	Should	Gold-type dropdown updates table data accordingly

M05 — Exchange Rate & Unit Converter — Tỷ giá ngoại tệ & Quy đổi

Provides USD/VND exchange rate and an online gold-value calculator.

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-05.1	System must display USD/VND rate updated at least every 15 minutes during banking hours	Must	<i>Rate timestamp is never older than 15 minutes; source displayed</i>
FR-05.2	System must provide a converter: user enters gold quantity (tael/oz) → returns VND/USD value	Must	<i>Result correct per formula; updates in real time as user types</i>
FR-05.3	System must support EUR/VND rate and conversion (R2)	Could	<i>R2: EUR/VND currency pair added; converter supports EUR output</i>

M06 — User Authentication — Xác thực tài khoản

Manages user registration, login, and account security.

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-06.1	System must allow Guests to register with email and password	Must	Account created with status = "pending"; verification email sent successfully
FR-06.2	Each email address may only be used for one account	Must	Duplicate email returns HTTP 409 with clear error message
FR-06.3	System must send email verification after registration; account activates only after verification	Must	Verification link valid for 24 h; expired link shows message and resend button
FR-06.4	Password must be at least 8 characters including at least 1 uppercase letter and 1 digit	Must	Client-side and server-side validation; specific error returned if policy not met
FR-06.5	System must issue JWT token (24 h) after successful login	Must	Valid token issued; refresh token stored in httpOnly cookie
FR-06.6	System must lock account for 15 minutes after 5 consecutive failed login attempts	Must	5th failure returns HTTP 429; remaining lockout time displayed
FR-06.7	System must support "Forgot Password" via email	Must	Reset link expires after 1 h; new password hashed with bcrypt before saving
FR-06.8	System must support Google OAuth 2.0 login (R2)	Could	R2: OAuth flow completes; new account created if email not found
FR-06.9	Logged-in users may change their password	Should	Old password validated; new password must meet policy
FR-06.10	Users may delete their account at any time; personal data purged within 30 days	Should	Soft delete applied; cannot log in after deletion; data clean after 30 days

M07 — Price Alerts — Cảnh báo giá

Allows Registered Users to set price threshold conditions and receive automatic email notifications (R2).

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-07.1	Registered Users must be able to create alerts: brand, gold type, threshold price, condition ($\leq$ or $\geq$)	Could	R2: Alert stored in DB with status = "active"; success confirmation shown
FR-07.2	Minimum alert threshold is 100,000 VND/tael	Could	Server-side validation; returns HTTP 400 if $< 100,000$ VND
FR-07.3	System must evaluate alert conditions after every price update cycle	Could	Background job runs after each crawl; checks all active alerts
FR-07.4	When condition is met, system must send email notification within 2 minutes	Could	Email reaches inbox within ≤ 2 minutes; content matches alert details
FR-07.5	System must automatically deactivate an alert after it fires (unless repeat mode is on)	Could	Status changes to "triggered"; no further email for that alert
FR-07.6	Minimum 30-minute cooldown between repeat notifications for the same condition	Could	Within 30 min after sending, no further email even if condition still met
FR-07.7	Each user may have at most 10 active alerts simultaneously	Could	11th creation attempt returns HTTP 400 with clear limit message
FR-07.8	Users must be able to view, edit, delete, and toggle each alert	Could	Full CRUD; toggle active/inactive without deleting
FR-07.9	Users may view a history of triggered alerts	Could	List shows trigger timestamp, price at trigger, send status
FR-07.10	System must support push notifications via iOS/Android app (R3)	Won't	Out of scope for R1–R2

M08 — Admin Dashboard — Quản trị hệ thống

Provides the system administrator with a full management interface.

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-08.1	Admin must have a dashboard showing: total users, today's visits, crawler status per source	Must	<i>Figures match DB; crawler status reflects actual log state</i>
FR-08.2	Admin must be able to add new data sources: name, URL, type (API/Crawl), frequency	Must	<i>New source saved; crawler applies configuration immediately</i>
FR-08.3	Admin must be able to edit and disable existing data sources	Must	<i>Disabling stops crawl immediately; last data still shown with Outdated status</i>
FR-08.4	Admin must be able to test a data source connection before saving	Must	<i>"Test connection" button returns OK or error with HTTP status code</i>
FR-08.5	System must log and display all anomaly-flagged data records	Must	<i>List shows: brand, gold type, old price, new price, % deviation, timestamp</i>
FR-08.6	Admin must be able to approve or reject anomalous data	Must	<i>Approve → record displayed; Reject → record permanently hidden</i>
FR-08.7	Admin must be able to view user list with basic information	Should	<i>Table shows: email, registration date, status, role</i>
FR-08.8	Admin must be able to lock / unlock user accounts	Must	<i>Locked user cannot log in; unlock takes effect immediately</i>
FR-08.9	Admin must be able to configure crawl frequency per source	Should	<i>Frequency change applies from next crawl cycle onward</i>
FR-08.10	Admin must be able to view system statistics reports by day/week/month	Could	<i>Report includes: visits, new users, alerts sent, crawl success rate</i>

PART 1B — CORE NON-FUNCTIONAL REQUIREMENTS (SRS v1.0)

4.1 Performance Requirements

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-P01	Performance	Home page load time ≤ 2 seconds on 4G/Wi-Fi	<i>Lighthouse Performance Score</i>
NFR-P02	Performance	Internal API response time ≤ 500 ms for 95% of requests	<i>Load test with k6 or JMeter</i>
NFR-P03	Performance	Support $\geq 1,000$ concurrent users at Release 1	<i>Stress test with 1,000 concurrent users</i>
NFR-P04	Performance	Price update latency ≤ 5 minutes vs source during trading hours	<i>Compare DB timestamp vs source timestamp</i>
NFR-P05	Performance	Alert email delivery ≤ 2 minutes from condition trigger	<i>Measure time from trigger to inbox arrival</i>

4.2 Reliability Requirements

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-R01	Reliability	System uptime $\geq 99.5\%$ per month (≤ 3.6 hours downtime/month)	<i>UptimeRobot continuous monitoring</i>
NFR-R02	Reliability	Data accuracy $\geq 99\%$ vs official sources on periodic checks	<i>Manual cross-check weekly</i>
NFR-R03	Reliability	Recovery Time Objective (RTO) ≤ 30 minutes	<i>Quarterly drill test</i>
NFR-R04	Reliability	Historical price data retained for a minimum of 5 years	<i>Periodic DB capacity check</i>

4.3 Security Requirements

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-S01	Security	Passwords hashed with bcrypt (salt rounds ≥ 10) before storage	<i>Code review; no plaintext password in DB</i>
NFR-S02	Security	All connections use HTTPS/TLS 1.2+; no plain HTTP endpoints	<i>SSL Labs scan; no HTTP redirect found</i>
NFR-S03	Security	JWT tokens expire in ≤ 24 h; refresh token stored in httpOnly cookie	<i>Token expiry test; cookie attribute check</i>

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-S04	Security	Brute-force protection: 15-minute lockout after 5 failed logins; rate limiting on all endpoints	<i>Automated login attempt test</i>
NFR-S05	Security	Personal data protection compliant with Decree 13/2023/NĐ-CP; data minimisation applied	<i>Privacy audit; only necessary fields collected</i>
NFR-S06	Security	All user inputs validated and sanitised; protection against SQL injection and XSS	<i>OWASP ZAP scan; penetration test</i>
NFR-S07	Security	Audit log records all Admin actions and sensitive data changes	<i>Log review; every admin action traceable</i>

4.4 Usability Requirements

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-U01	Usability	Responsive design: works correctly on screens ≥ 320 px; no layout breakage	<i>Cross-device testing: 320 px, 768 px, 1440 px, 2560 px</i>
NFR-U02	Usability	Compatible with Chrome, Firefox, and Safari (2 latest versions each)	<i>Browser compatibility matrix test</i>
NFR-U03	Usability	New users can look up a gold price within ≤ 5 minutes without documentation	<i>Usability test with 5 new participants</i>
NFR-U04	Usability	Error messages written in clear Vietnamese; no stack traces exposed to users	<i>Error message review; no technical jargon visible</i>
NFR-U05	Usability	Loading states shown (skeleton / spinner) while data is fetching; no blank screen	<i>Throttle network to 3G; verify loading indicator always present</i>
NFR-U06	Usability	Keyboard navigation supported for main components; contrast ratio $\geq 4.5:1$	<i>axe accessibility audit; keyboard-only navigation test</i>

4.5 Scalability Requirements

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-SC01	Scalability	Architecture allows adding new crawlers without refactoring the core module	<i>Code review; new crawler added without touching core logic</i>
NFR-SC02	Scalability	System supports auto-scaling on traffic spikes (AWS Auto Scaling)	<i>Load spike simulation; verify instances scale within 2 minutes</i>
NFR-	Scalability	Database supports read replicas; Redis	<i>Replica lag ≤ 1 second under</i>

Mã NFR	Loại	Yêu cầu	Cách đo lường
SC03		cluster when needed	<i>normal load</i>

4.6 Maintainability Requirements

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-M01	Maintainability	Unit test coverage $\geq 70\%$ for business logic; CI/CD runs tests before every deploy	<i>Coverage report from Jest/pytest; CI pipeline enforces threshold</i>
NFR-M02	Maintainability	Swagger/OpenAPI documentation complete and updated in parallel with code	<i>Docs coverage check; all endpoints documented</i>
NFR-M03	Maintainability	Centralised logging: application log + error log + crawler log; configurable log levels	<i>Log aggregation tool (e.g. Loki) ingests all three streams</i>
NFR-M04	Maintainability	Lint check mandatory (ESLint/Pylint); no lint errors before merge	<i>CI pipeline blocks merge if lint fails</i>
NFR-M05	Maintainability	Package lock file committed to repo; outdated dependencies reviewed periodically	<i>Dependabot or Renovate configured; review schedule documented</i>

PART 1C — AUTHORIZATION MATRIX

Feature / Functionality	Guest	Registered User	Admin
View domestic gold prices	✓	✓	✓
View international gold prices	✓	✓	✓
View price history charts	✓	✓	✓
View brand comparison table	✓	✓	✓
View exchange rates	✓	✓	✓
Register / Log in	✓	—	✓
Create and manage price alerts	✗	✓	✓
Save personal settings	✗	✓	✓
Export history to CSV	✗	✓ (R3)	✓
Manage data sources	✗	✗	✓
Add / Edit / Remove gold brands	✗	✗	✓
View system logs and anomaly data	✗	✗	✓
Manage user accounts	✗	✗	✓
Configure crawl frequency	✗	✗	✓
View system statistics reports	✗	✗	✓

✓ = Access granted | ✗ = No access | — = Not applicable

PART 2 — NEW FEATURES (Addendum v2.0 — F01–F11)

These 11 features are proposed additions that differentiate GPLS from existing Vietnamese gold price websites. They build on top of the core SRS v1.0 modules and are targeted across R1, R2, and R3.

Overview — All 11 New Features

ID	Feature Name	One-line Description	Target Users	Release	Complexity
F01	 AI Gold Assistant	Integrates an AI chatbot (OpenAI GPT-4o / Gemini) enabling users to ask natural-language questions about the gold market.	Reg. User + Guest	R2	Low
F02	 Gold Portfolio Tracker	Lets users log personal gold buy/sell transactions.	Reg. User	R2	Med
F03	 Market Heat Index	A 0–100 index calculated from price velocity, buy-sell spread size, and number of threshold crossings during the day.	All Users	R1	Low
F04	 Morning Gold Digest	Every morning at 07:30 the system auto-generates a short market summary using AI or a smart template, covering overnight movements, weekly comparison, and highlights.	Reg. User (opt-in)	R2	Med
F05	 Smart Alert Engine	Extends the basic alert system (FR07) with multi-condition logic: trend-based alerts (price drops N times in a row), spread-based alerts (buy-sell gap narrows below threshold), and AND combinations of two conditions..	Reg. User	R2	Med
F06	 DCA Simulator	Simulates a Dollar-Cost Averaging strategy using real internal historical data.	All Users	R1	Low
F07	 Community Price Forecast	Each day Registered Users vote on tomorrow's SJC price direction: Up / Down / Flat.	Reg. User	R3	High

ID	Feature Name	One-line Description	Target Users	Release	Complexity
F08	 Advanced Unit Converter	Extends the basic converter (FR05.2) to support 6 weight units (Tael/Chi/Phân/Troy oz/Gram/Kg), cross-referenced with live gold prices in VND/USD/EUR and selectable gold purity (24K/22K/18K/14K)..	All Users	R1	Low
F09	 Personalized Dashboard	The system learns which brand-gold type combinations each user views most frequently and automatically reorders the home page price table accordingly.	Reg. User	R2	Med
F10	 Browsing History & Smart Suggestions	Logs which prices each user has viewed and when.	Reg. User	R2	Med
F11	 Spread Comparison Dashboard	Visualises the buy-sell spread of every active brand as a ranked horizontal bar chart.	All Users	R1	Low

F01 — 🤖 AI Gold Assistant — Trợ lý phân tích vàng thông minh

Overview

Integrates an AI chatbot (OpenAI GPT-4o / Gemini) enabling users to ask natural-language questions about the gold market. The AI is provided with real-time price context from GPLS and can analyse trends, explain movements, and compare brands on demand.

🗨️ *Differentiator: No gold price website in Vietnam currently has an AI assistant integrated with live data.*

Release Target	R2
Target Users	Registered User + Guest
Related Screens	Floating chat widget (all pages) Dedicated AI Insights page (Registered User)

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F01.1	System must provide an AI chatbot widget on the web interface	Could	Chat widget visible bottom-right; user types freely; AI responds in ≤ 5 s
FR-F01.2	AI must be supplied with a real-time gold price snapshot before answering	Could	System prompt includes current SJC, DOJI, XAU/USD prices; answers reflect live data
FR-F01.3	AI must decline questions outside the gold market scope	Could	Off-topic queries receive a polite refusal; no hallucinated financial data
FR-F01.4	AI must append "For reference only" disclaimer to every answer containing price values	Could	Disclaimer always present at end of response; user cannot suppress it
FR-F01.5	Guests limited to 10 questions/day; Registered Users unlimited	Could	Counter tracked in session/DB; Guest exceeding limit sees account-registration prompt
FR-F01.6	System must display 3–5 suggested questions when chat is first opened	Could	Suggestion chips auto-fill input on click; refreshed daily

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F01.1	Performance	AI response P90 latency $\leq$ 5 seconds	<i>APM tool; alert if P90 exceeds 8 s</i>
NFR-F01.2	Cost	AI API cost $\leq$ \$150/month at 10,000 questions/month	<i>OpenAI usage dashboard weekly review</i>
NFR-F01.3	Privacy	No user PII passed in AI prompts	<i>Code review: prompt contains only market data, no user identifiers</i>

F02 — Gold Portfolio Tracker — Sổ theo dõi danh mục vàng cá nhân

Overview

Lets users log personal gold buy/sell transactions. The system automatically calculates P&L against current live prices and displays a portfolio value chart and allocation breakdown over time.

 *Differentiator: Transforms GPLS from a lookup tool into a personal gold asset manager — the single strongest reason to register and return daily.*

Release Target	R2
Target Users	Registered User
Related Screens	Portfolio page (Registered User) Add/Edit transaction form Portfolio overview dashboard

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F02.1	Registered Users must be able to add transactions: type (Buy/Sell), brand, gold type, quantity (tael/chỉ), price, date	Could	Form with validation (quantity > 0, date ≤ today); transaction saved and listed
FR-F02.2	System must auto-calculate current portfolio value using live prices, refreshed every 5 minutes	Could	Value = $\Sigma(\text{quantity} \times \text{current buy price for matching brand})$; updates each crawl cycle
FR-F02.3	System must display P&L (VND amount and %) per position and for total portfolio	Could	$P\&L = (\text{currentPrice} - \text{costPrice}) \times \text{qty}$; green/red colouring; live updates
FR-F02.4	System must display a P&L line chart over time for the full portfolio	Could	Line chart from first transaction date to today; hover tooltip shows value on any date
FR-F02.5	System must display allocation breakdown by brand and gold type (pie chart + table)	Could	Chart and percentage table update whenever a transaction is added or edited
FR-F02.6	Registered Users must be able to edit and delete individual transactions	Could	Edit/Delete on each row; confirm dialog before delete; P&L recalculated immediately

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F02.1	Privacy	Portfolio data is visible only to the owning user; Admin cannot see transaction content	<i>Penetration test; API authorisation check</i>
NFR-F02.2	Data Integrity	Portfolio data is preserved when user changes email or password	<i>Test: update account info → verify portfolio intact</i>

F03 — 📌 Market Heat Index — Chỉ số nhiệt độ thị trường

Overview

A 0–100 index calculated from price velocity, buy-sell spread size, and number of threshold crossings during the day. Displayed as a gauge chart labelled Cold / Warm / Hot. Fully computed from internal data — no external AI API required.

🕒 *Differentiator: Users understand market activity in 3 seconds — no Vietnamese gold site provides this at-a-glance signal.*

Release Target	R1
Target Users	All Users
Related Screens	Gauge widget on home page Market Heat detail page (R2)

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F03.1	System must compute Market Heat Index (0–100) after every price update	Should	<i>Index = price velocity (40%) + spread size (30%) + threshold crossings today (30%)</i>
FR-F03.2	System must classify Index into Cold (0–33, blue), Warm (34–66, yellow), Hot (67–100, red)	Should	<i>Colour and label match thresholds; auto-updates every 5 minutes</i>
FR-F03.3	System must render a gauge chart on the home page	Should	<i>Gauge needle points to current value; updates every 5 minutes</i>
FR-F03.4	Hovering/clicking the gauge must show a tooltip explaining the three components	Should	<i>Tooltip shows: price velocity X%, spread Y VND, N threshold crossings today</i>
FR-F03.5	System must store hourly Heat Index history for a 7-day trend chart (R2)	Could	<i>R2: line chart of index over 7 days; admin can view full history</i>

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F03.1	Performance	Index calculation completes within ≤ 100 ms after each crawl cycle	<i>Background job timing log; alert if > 500 ms</i>

F04 — Morning Gold Digest — Bản tin thị trường tự động mỗi sáng

Overview

Every morning at 07:30 the system auto-generates a short market summary using AI or a smart template, covering overnight movements, weekly comparison, and highlights. Delivered by email (opt-in) and displayed as a card on the home page after login.

 *Differentiator: Creates the habit of "opening GPLS every morning" — a powerful retention tool that no Vietnamese gold site provides.*

Release Target	R2
Target Users	Registered User (opt-in)
Related Screens	Digest card on home page (Registered User) Digest archive page /digest/archive Settings: toggle email delivery

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F04.1	System must auto-generate Morning Digest at 07:30 on every business day	Could	<i>Cron job runs on schedule; digest generated within ≤ 2 min; fallback template if AI unavailable</i>
FR-F04.2	Digest must include: current SJC/DOJI prices, % change vs yesterday, XAU/USD, and one notable highlight	Could	<i>All four sections present; figures correct per DB; AI adds 2–3 sentence analysis if available</i>
FR-F04.3	Registered Users must be able to opt in/out of email delivery	Could	<i>Toggle in Profile Settings; change takes effect from the following day</i>
FR-F04.4	Digest card must appear on the home page for logged-in users after login	Could	<i>"Today's Digest" banner shown above price table; user can dismiss it</i>
FR-F04.5	System must store digest history and allow retrieval of past digests	Could	<i>Page /digest/archive lists digests by date; any past digest viewable</i>

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F04.1	Reliability	Digest generated successfully on $\geq 99\%$ of business days	<i>Admin alert if generation fails 2 consecutive days</i>
NFR-F04.2	Cost	Each digest consumes ≤ 500 AI tokens (prompt optimised)	<i>Token usage log in Admin dashboard</i>

F05 — Smart Alert Engine — Cảnh báo thông minh nâng cao

Overview

Extends the basic alert system (FR07) with multi-condition logic: trend-based alerts (price drops N times in a row), spread-based alerts (buy-sell gap narrows below threshold), and AND combinations of two conditions.

 *Differentiator: The base alert system supports only a fixed price threshold. Smart Alerts enable professional-grade strategies not found on any Vietnamese gold site.*

Release Target	R2
Target Users	Registered User
Related Screens	Advanced alert builder UI Alert list (updated UI)

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F05.1	System must support trend alerts: price rises or falls N consecutive times (N = 2–5)	Could	Builder has "Trend Alert" option; N selectable from 2–5; engine checks pattern across N recent crawl cycles
FR-F05.2	System must support spread alerts: buy-sell gap $\leq$ X VND	Could	"Spread Alert" option in builder; X validated ≥ 0 ; engine calculates spread = sellPrice – buyPrice each cycle
FR-F05.3	System must support combining two conditions with AND logic	Could	UI allows Condition 1 AND Condition 2; both must be true simultaneously before alert fires
FR-F05.4	System must display the combined condition as natural-language Vietnamese text	Could	"Alert me when SJC drops 3 times in a row AND spread is below 200,000 VND" — generated automatically from selections
FR-F05.5	Smart Alert email must include a 24-hour price chart image	Could	HTML email contains server-generated PNG chart; trigger timestamp and price clearly stated

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F05.1	Performance	Alert evaluation engine processes all Smart Alerts within ≤ 500 ms after each crawl cycle	<i>Job timing log; alert if > 1 second</i>

F06 — □ DCA Simulator — Máy tính chiến lược mua vàng đều đặn

Overview

Simulates a Dollar-Cost Averaging strategy using real internal historical data. Users choose a brand, gold type, period, frequency (weekly/monthly), and quantity per purchase. The system calculates average cost, total holdings, and compares DCA vs lump-sum with a dual-line chart.

 *Differentiator: Educates users to make evidence-based decisions using historical data already in the DB — no external API required.*

Release Target	R1
Target Users	All Users
Related Screens	Page /tools/dca-simulator Simulation results with comparative chart

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F06.1	DCA Simulator must be accessible without login	Should	Page /tools/dca-simulator; no login required; fully responsive
FR-F06.2	Simulator must accept: brand, gold type, start date, frequency (weekly/monthly), quantity per purchase	Should	Form validation: period $\leq$ available history; quantity > 0 ; results appear within ≤ 1 second
FR-F06.3	System must calculate and display: average cost, total gold, total spent, current value, P&L vs lump-sum	Should	All 5 metrics shown with brief explanation; figures match DB data
FR-F06.4	System must display a dual-line chart comparing DCA vs Lump Sum over the selected period	Should	Two labelled lines; hover tooltip at each purchase point
FR-F06.5	Registered Users may save simulation results to their Portfolio Tracker	Could	"Save to Portfolio" button; DCA transactions auto-populated in F02

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F06.1	Performance	DCA calculation for 2 years $\times$ weekly frequency (≈ 104 data points) completes	Client-side timing; tested with real dataset

Mã NFR	Loại	Yêu cầu	Cách đo lường
		within $\leq$ 1 second	

F07 — Community Price Forecast — Dự đoán giá cộng đồng

Overview

Each day Registered Users vote on tomorrow's SJC price direction: Up / Down / Flat. Public sentiment ratio visible after voting. System scores predictions the next morning based on actual price. Monthly leaderboard drives community engagement.

 *Differentiator: Turns price tracking into a daily social activity — increasing session frequency and community-driven virality.*

Release Target	R3
Target Users	Registered User
Related Screens	Vote widget on home page Leaderboard page Personal prediction history

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F07.1	System must open the voting session at 17:00 and close it at 07:00 the following day	Could	<i>Cron job manages session state; outside window shows "Voting closed"</i>
FR-F07.2	Registered Users may vote once per day; vote may be changed before session closes	Could	<i>DB stores (user_id, date); UPDATE if already exists; voting blocked when session closed</i>
FR-F07.3	Vote ratio must be visible to all users after the user has voted	Could	<i>Before voting: ratio hidden; after voting: % Up/Down/Flat shown; Guests can see after 12:00 daily</i>
FR-F07.4	System must score predictions at 09:00 the next day based on actual SJC price	Could	<i>Up = opening > previous close; correct prediction = +10 pts; wrong = 0 pts</i>
FR-F07.5	System must maintain a monthly leaderboard of top 20 most accurate predictors	Could	<i>Leaderboard updates daily; shows display name, points, correct-prediction streak</i>

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F07.1	Fairness	Vote records are immutable once the session closes; Admin cannot retroactively alter results	<i>Audit log; DB constraint prevents UPDATE after session_closed = true</i>

F08 — ⚖️ Advanced Unit Converter — Bộ chuyển đổi đơn vị toàn diện

Overview

Extends the basic converter (FR05.2) to support 6 weight units (Tael/Chỉ/Phân/Troy oz/Gram/Kg), cross-referenced with live gold prices in VND/USD/EUR and selectable gold purity (24K/22K/18K/14K).

 *Differentiator: Traditional Vietnamese gold units (Tael/Chỉ/Phân) are absent from any popular converter — this fills a real gap for jewellery buyers.*

Release Target	R1
Target Users	All Users
Related Screens	Page /tools/converter (extends FR05.2) Embedded converter widget on exchange rate page

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F08.1	Converter must support 6 weight units: Tael (Lượng), Chỉ, Phân, Troy Ounce, Gram, Kilogram	Should	6-unit dropdown; typing in any field updates all others in real time; conversion constants accurate to 6 decimal places
FR-F08.2	Converter must calculate monetary value by gold purity: 24K, 22K, 18K, 14K	Should	Purity selector; value = $(\text{weight/standard unit}) \times \text{price/tael} \times (\text{purity}/24)$; uses live DB price
FR-F08.3	Converter must display results in three currencies simultaneously: VND, USD, EUR	Should	Three result columns updated in real time using live exchange rates from DB
FR-F08.4	Converter must have a "Copy result" button per row	Could	Click copies text like "3 Chỉ 18K gold = 15,750,000 VND (SJC price at 09:15 01/06/2026)"; toast confirmation shown

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F08.1	Accuracy	Conversion constants accurate to 6 decimal places	<i>Unit test: 1 Tael = 37.500000 g; 1 Troy oz = 31.103477 g</i>

F09 — Personalized Dashboard — Giao diện cá nhân hóa theo thói quen

Overview

The system learns which brand-gold type combinations each user views most frequently and automatically reorders the home page price table accordingly. Users can also manually pin up to 5 favourite pairs. Preferences sync across devices server-side.

 *Differentiator: Every Vietnamese gold site shows the same static table to every user. GPLS is the first to "know who you are" and personalise the interface accordingly.*

Release Target	R2
Target Users	Registered User
Related Screens	Home page (reordered price table) Settings → Personalisation "Your Favourites" widget

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F09.1	System must asynchronously record view frequency per (brand, gold type) pair for each Registered User	Could	Every price row click/load → async DB event write; non-blocking; no impact on API response time
FR-F09.2	System must reorder the price table to show the top 3 most-viewed pairs first	Could	After 3 login sessions with data, table auto-reorders; "Personalised for you" badge shown
FR-F09.3	Registered Users must be able to pin up to 5 (brand, gold type) pairs to the top of the table	Could	Pin icon on each row; pinned items always appear first with pin icon; drag-and-drop to reorder pins
FR-F09.4	System must allow users to reset to default order at any time	Could	"Restore default" button in Settings; deletes all preferences and pins after confirm dialog
FR-F09.5	Personalisation preferences must sync across devices upon login	Could	Preferences stored server-side by user_id; no localStorage; synced immediately on login from any device
FR-F09.6	Guests see the default table order and are prompted to register to personalise	Could	No pin icon for Guests; banner "Log in to personalise your view" shown

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F09.1	Performance	Behavioural tracking must not increase price-table API response time by more than 50 ms	<i>Fire-and-forget async write; measure P95 before and after enabling tracking</i>
NFR-F09.2	Privacy	Behavioural data used only for personalisation; never shared or sold to third parties	<i>Data flow diagram confirms no external calls containing user behavioural data</i>
NFR-F09.3	Storage	Behavioural event data retained for a maximum of 90 days; older records deleted by daily cron job	<i>DB audit: no behavioural_event record older than 90 days after cleanup runs</i>

F10 — 🕒 Browsing History & Smart Suggestions — Lịch sử tra cứu & Gợi ý thông minh

Overview

Logs which prices each user has viewed and when. Displays inline context beneath current prices: "Last viewed 12 days ago at 85,500,000 VND — today is +2.3%". Includes a personal lowest-price tracker to support purchase timing decisions.

🔗 *Differentiator: Turns every lookup into a connected journey — users see their personal context, not just an isolated price. No Vietnamese gold site offers this.*

Release Target	R2
Target Users	Registered User
Related Screens	Inline suggestion beneath price row on main table Page /profile/history "Recently Viewed" widget on home page

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F10.1	System must log Registered User price views: brand, gold type, price at view time, timestamp	Could	Click on any price row → record in view_history table; max 500 most recent records per user
FR-F10.2	System must show inline context beneath the current price: "Last viewed: X days ago at Y VND (±Z%)"	Could	Sub-text or tooltip below current price; only shown for pairs viewed at least once; % = $(\text{currentPrice} - \text{histPrice}) / \text{histPrice} \times 100$
FR-F10.3	Registered Users must be able to view a full browsing history page	Could	Page /profile/history; sorted newest first; filterable by brand/gold type; paginated 20 items/page
FR-F10.4	System must display "Lowest price you have ever seen" for each viewed (brand, gold type) pair	Could	Dedicated column in history page; compare vs current price; highlight green if today is a new personal low
FR-F10.5	Registered Users must be able to clear their entire browsing history	Could	"Clear all history" button in Settings; confirm dialog; all inline suggestions disappear immediately after

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F10.1	Privacy	Browsing history completely purged within 30 days of account deletion (compliant with Decree 13/2023)	<i>Test: delete account → verify view_history table is empty after 30 days</i>
NFR-F10.2	Performance	History page API responds within ≤ 800 ms for users with 500 records	<i>Load test; DB indexed on (user_id, viewed_at) DESC</i>

F11 — Spread Comparison Dashboard — Bảng so sánh chênh lệch mua-bán trực quan

Overview

Visualises the buy-sell spread of every active brand as a ranked horizontal bar chart. The brand with the smallest spread is automatically labelled "Most cost-efficient" — the most useful signal for short-term traders. Includes a plain-language explanation for non-expert users.

 *Differentiator: FR04 already shows a price comparison table, but does not treat spread as a standalone metric. Spread is the most important cost signal for short-term gold buyers — and no Vietnamese site presents it clearly.*

Release Target	R1
Target Users	All Users
Related Screens	Spread section expanding FR04 on the comparison page Dedicated page /spread-ranking Spread mini-widget on home page

Functional Requirements (FR)

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F11.1	System must calculate and display spread (VND and %) per brand in real time	Should	$Spread\ VND = sellPrice - buyPrice$; $Spread\ \% = (sellPrice - buyPrice) / buyPrice \times 100$; refreshed every 5 minutes
FR-F11.2	System must rank brands by spread ascending; top brand automatically labelled "Most cost-efficient"	Should	Table sorted by spread ascending; top row highlighted green + badge "Most cost-efficient for short-term trading"; tooltip explains
FR-F11.3	System must display a horizontal bar chart comparing brand spreads simultaneously	Should	Bar chart: Y-axis = brand name, X-axis = spread VND; green-yellow-red gradient by spread size; renders within ≤ 1 second
FR-F11.4	Users must be able to filter spread view by gold type	Should	Gold-type dropdown; table and chart reload with correct data
FR-F11.5	System must display a plain-language explanation of "spread" accessible via info icon	Should	"?" icon next to "Spread" heading; tooltip reads: "Spread is how much you lose if you buy and sell immediately. Smaller spread = less cost."

Mã FR	Yêu cầu	Ưu tiên	Acceptance Criteria
FR-F11.6	System must store spread history and show a 7-day trend chart per brand (R2)	Could	<i>R2: hourly spread line chart for 7 days; hover tooltip on any point; data sourced from existing price_history table</i>

Non-Functional Requirements (NFR)

Mã NFR	Loại	Yêu cầu	Cách đo lường
NFR-F11.1	Accuracy	Spread must only be calculated from buyPrice and sellPrice sharing the same crawl_session_id	<i>Unit test: spread = null when buy and sell come from different sessions; DB constraint enforced</i>
NFR-F11.2	Performance	Spread Dashboard including bar chart renders within ≤ 2 seconds	<i>Lighthouse test; lazy-load chart if necessary</i>

PART 3 — RELEASE ROADMAP & IMPACT MATRIX

Release Schedule — Combined Core + New Features

Release	Deliverables	Goal
R1 — Q2/2026	Core: M01 M02 M03 M04 M05 M06 (partial) M08 New: F03 Market Heat Index · F06 DCA Simulator · F08 Advanced Converter · F11 Spread Dashboard	Full core lookup platform live; 4 differentiating tools launched
R2 — Q3/2026	Core: M06 (OAuth) · M07 Price Alerts New: F01 AI Assistant · F02 Portfolio Tracker · F04 Morning Digest · F05 Smart Alert Engine · F09 Personalized Dashboard · F10 Browsing History	Strong personalisation layer; daily return incentive; registered-user value maximised
R3 — Q4/2026	Core: M03 (3M/1Y charts) · M04 (all brands) · CSV export New: F07 Community Forecast + Leaderboard	Community features; full brand coverage; viral growth mechanics

Impact Matrix — New Features vs Business Goals

Feature	Signups ↑	Retention ↑	Differentiation	Revenue	SEO	Notes
F01 AI Assistant	☆☆	☆☆☆	☆☆☆	☆☆	☆	API cost needs tight control
F02 Portfolio Tracker	☆☆☆	☆☆☆	☆☆☆	☆☆☆	☆☆	Highest overall impact
F03 Heat Index	☆	☆☆	☆☆☆	☆	☆☆	R1, zero external cost
F04 Morning Digest	☆	☆☆☆	☆☆	☆	☆☆☆	Email habit = SEO + retention
F05 Smart Alert	☆☆	☆☆☆	☆☆☆	☆☆	☆	Builds on core FR07
F06 DCA Simulator	☆	☆☆	☆☆☆	☆	☆☆☆	Tool content = strong SEO
F07 Community	☆☆☆	☆☆☆	☆☆☆	☆☆	☆☆	R3 — needs user base first
F08 Converter	☆	☆	☆☆	☆	☆☆☆	R1, easy

Feature	Signups ↑	Retention ↑	Differentiation	Revenue	SEO	Notes
						build, SEO value
F09 Personalised UI	☆☆☆	☆☆☆	☆☆☆	☆☆	☆	Strong registration incentive
F10 History & Suggest	☆☆	☆☆☆	☆☆☆	☆	☆☆	Personal context = loyalty
F11 Spread Dashboard	☆☆	☆☆	☆☆☆	☆	☆☆☆	R1, unique, no competitor has it

Recommended implementation order: F11 → F08 → F03 → F06 (R1: quick wins, zero AI cost) → F02 → F05 → F09 → F10 → F04 → F01 (R2: core personalisation value) → F07 (R3: community)